

Introduction to Derivatives

- **Definition:** A financial instrument whose value is derived from the value of an underlying asset.
- **Underlying Assets:** Can be commodities (gold, oil), currencies (USD, EUR), stocks, bonds, interest rates, market indices (Nifty, Sensex), etc.
- **Purpose:**
 - **Hedging:** To reduce risk by taking an offsetting position.
 - **Speculation:** To profit from anticipated price movements of the underlying asset.
 - **Arbitrage:** To profit from price discrepancies in different markets.
- **Key Characteristics:**
 1. Value derived from an underlying asset.
 2. Requires little or no initial net investment (compared to the notional amount).
 3. Settled at a future date.

Types of Derivatives

There are four main types of derivative instruments:

1. Forwards

- **Definition:** A customized contract between two parties to buy or sell an asset at a specified price on a future date.
- **Key Features:**
 - Over-the-Counter (OTC) market.
 - Highly customizable (asset, quantity, date).
 - Higher default risk (counterparty risk).
 - No intermediate cash flows (usually).

2. Futures

- **Definition:** A standardized contract between two parties to buy or sell an asset at a specified price on a future date, traded on an exchange.
- **Key Features:**
 - Exchange-traded.
 - Standardized (contract size, expiry date, quality).
 - Reduced default risk due to clearing house.
 - Marked-to-market daily (daily settlement of gains/losses).

3. Options

- **Definition:** A contract that gives the buyer the right, but not the obligation, to buy or sell an underlying asset at a specified price (strike price) on or before a specified date (expiry date).
- **Types:**
 - **Call Option:** Gives the holder the right to buy the underlying asset.
 - **Put Option:** Gives the holder the right to sell the underlying asset.
- **Key Terms:**
 - **Premium:** Price paid by the buyer to the seller for the option.
 - **Strike Price:** The predetermined price at which the underlying asset can be bought or sold.
 - **Expiry Date:** The last date on which the option can be exercised.
- **American vs. European Options:**
 - **American:** Can be exercised any time up to the expiry date.
 - **European:** Can only be exercised on the expiry date.

4. Swaps

- **Definition:** An agreement between two parties to exchange future cash flows based on different underlying assets or rates.
- **Common Types:**
 - **Interest Rate Swap:** Exchange fixed interest payments for floating interest payments.
 - **Currency Swap:** Exchange principal and/or interest payments in one currency for equivalent payments in another currency.
- **Key Features:**
 - OTC market.
 - Customizable.
 - Used for hedging interest rate or currency risk.

Participants in Derivative Markets

1. **Hedgers:** Use derivatives to reduce or eliminate risk exposure from underlying assets.
2. **Speculators:** Take positions to profit from anticipated price movements of the underlying asset. They take on risk hoping for high returns.
3. **Arbitrageurs:** Seek to profit from price discrepancies of the same asset in different markets. They aim for risk-free profit.

Advantages and Disadvantages

Advantages

- **Risk Management:** Effective tool for hedging against various financial risks.
- **Price Discovery:** Help in efficient price discovery of the underlying assets.
- **Market Efficiency:** Facilitate quicker and cheaper transactions.
- **Leverage:** Can provide high returns with relatively small initial investment.

Disadvantages

- **High Risk (for Speculators):** Can lead to significant losses due to high leverage.
- **Complexity:** Can be complex and difficult to understand.
- **Counterparty Risk:** Especially in OTC derivatives like forwards and swaps.
- **Market Volatility:** Can amplify market volatility if misused.